

HERU PURNAMA

Research and Development Engineer | Network & System Engineer

Bandung | herupurnama.com | herupurnama.work@gmail.com | linkedin.com/in/heru-purnama
herupurnama.com/HeruPurnama_Portfolio.pdf

Versatile IT and R&D Engineer with experience in full-stack development, network and system engineering, and IT GRC. Experienced in troubleshooting, automation, infrastructure optimization, and developing secure web and IoT solutions. Passionate about research-driven problem solving, system improvement, and cross-functional product development.

EDUCATION

Universitas Pendidikan Indonesia

Electrical Engineering

2019 – 2023

- Actively participating in the Kampus Merdeka program.
- Engaged in server management for ee.upi.edu, involving routers, switches, firewalls, NAT, and port forwarding, Proxmox.
- Contribute to Faculty Research project: SmartVest, an assistive navigation for visually impaired in charge for GPS system using SIM800L communication and integrate with Telegram.
- Contribute to Faculty Research project: Smart Parking System using Microcontroller and API integration.
- Developed a website to provide easy access to the TOPSIS algorithm, using Flask and Python as a contribution for Solar Panel research.
- Engaging in off-campus activities by enrolling in various online courses and webinars.
- Experience in operating AutoCAD, SketchUp, and Blender for building and electrical drawings and 3D Design.

SMKN 5 Bandung

Computer and Network Engineering

2016 – 2019

- Actively involved in IT Club organization.
- Experienced in various school projects involving configuration of MikroTik routers, 3D modeling using Blender, game development in Unity, creating virtual lab environments using Linux, and developing basic CRUD systems using PHP CodeIgniter.

WORKING EXPERIENCE

Information Technology

Cikarang

LGE Indonesia Research and Development Center

Dec 2025 – Present

- Managed lifecycle, compliance, and deployment strategy for 600+ enterprise devices, improving asset traceability and operational readiness.
- Administered and monitored 140+ CCTV units, improving surveillance reliability and incident response.
- Managed procurement processes for IT infrastructure, including CCTV systems and access control equipment.
- Conducted internal bidding and vendor selection to ensure cost efficiency and compliance.
- Maintained and monitored IT asset inventory to support operational needs.
- Implemented and managed security solutions such as Data Loss Prevention (DLP), Endpoint Security, Network Access Control (NAC), and Wireless ISP (WISP).
- Integrated Cloud Extreme Controller API with LGE internal portal to automate network management processes.
- Developed an automated monitoring system for CCTV and access control equipment to improve system reliability and response time.
- Controlling External Storage (Flash Drive, HDD, SSD) through full procurement lifecycle.
- Conduct Training on Cyber Security to Researcher and Worker: Software Supply Chain Risk, Phishing Awareness, Man in the Middle Awareness, Rogue Access Point, Introducing on similar character but different ASCII code (bca.co.id and bca.co.id).
- Managing In/Out Employee from Training New Employee, Giving Access until Revoking Access.

Network and System Engineer

Bandung

Teradata Megah

June 2024 – Dec 2025

Working at an IT solutions company serving the finance sector as part of a four-person team, supporting 40 internal users (including programmers and business support staff) and 6 banking clients. I coordinate with various partners, including ISPs and payment gateway providers, and am responsible for managing the Network and System Infrastructure, ensuring the Data Centre operates reliably 24/7. My responsibilities include:

- Managing virtualization environments using VMware and Citrix.
- Implementing load balancing with Nginx and Citrix to optimize and distribute traffic efficiently.
- Database Replication, to ensure high availability data.
- Automating IPsec and OpenVPN tunnels for secure, rapid site-to-site deployments, integrated with Policy-Based Routing (PBR) and Network Address Translation (NAT).
- Administering network devices including Checkpoint, Mikrotik, and Aruba for security, routing, and wireless connectivity.

- Operating Extended Detection Response: XDR Bitdefender, EDR Eset to enhance endpoint security.

ACTIVITIES

Scalable Web Service With GoLang

Hacktiv8

<https://www.herupurnama.com/projects/2-GoLang-Project>

Indonesia, Remote

Feb 2024 - Mar 2024

Built and deployed scalable web services using Go (Golang) as part of a backend development program. Key projects include:

- Developing code to import data from Excel.
- Creating a REST API, and designing a minimal web game similar to Fallout Shelter.
- Creating MyGram API, which includes user management (login and register), photo uploads, commenting functionality, and social media interactions (post, get, put, delete) related to userIDs, and deploying it to Railway.

Artificial Intelligence Mastery Program

Orbit Future Academy

<https://www.herupurnama.com/shortcut/relate-project>

RELATE (Repair Electronic Website) – Group Project Lead

Indonesia, Remote

Feb 2022 - Jul 2022

A group project to develop a server-side rendered (SSR) web application that helps users identify their device's screen condition and receive service recommendations using custom-trained Machine Learning models. I led a cross-functional team with expertise in Machine Learning (ML), Natural Language Processing (NLP), Computer Vision (CV), and Data Science (DS), while also contributing as a Full Stack Developer using Python and Django. Key projects include

- Implemented Git and Trello to streamline team collaboration and agile workflow.
- Integrated an NLP chatbot that recommends relevant YouTube videos and Google Maps locations based on user queries.
- Integrated Computer Vision to classify smartphone screen damage into categories (e.g., minor, moderate, severe).
- Deployed the application on Heroku and a private server, supporting features like authentication, CRUD operations, Google APIs, and a user profile system.

Computer Network Engineer & IT Helpdesk

DitSTI ITB

Bandung, Indonesia

Jul 2018 – Sep 2018

- Offer client services, including hardware upgrades and repairs (primarily laptops), installing, licensing, and repairing software to various operating systems.
- LAN installation in new building and made perfect server with Control Panel using ISPConfig.

PROJECTS

Smart Home Automation & SmartVest Telegram API Integration

<https://www.herupurnama.com/shortcut/smart-home>

Apr 2023 - May 2023

Developed an IoT monitoring system using ESP32 and MQTT for real-time data transmission. A Python Flask server acts as a middleware bridge between the SIM800L module and external services due to TLS version limitations on the SIM800L. The server receives sensor and GPS data through lightweight GET/POST API endpoints, processes the data, stores it in a Supabase database, and forwards notifications to Telegram via webhook integration.

Example API endpoint:

/api/koordinat?lat=x&long=y&battery=z&altitude=a

Security Method Analysis on MikroTik RouterOS

<https://www.herupurnama.com/shortcut/security-analysis>

August 2023

Researched MikroTik RouterOS security filtering performance under DDoS attacks using TOPSIS and entropy-weight analysis. Compared filter drop, tarpit, reject, and raw drop methods based on CPU, memory, and traffic metrics, concluding that Raw Drop provided the most efficient mitigation performance.

VOLUNTEER EXPERIENCE

KKN Community Service

SDN Cibanteng & SMPN 59

<https://www.herupurnama.com/shortcut/kkn>

Bandung, Indonesia

January 2023

Led a multidisciplinary team to promote digital literacy in the community, focusing on cybersecurity and the use of technology. A key highlight was my article on Kompasiana, titled "Cara Mengamankan Diri di Dunia Digital," which received over 28,989 views. I also took initiative to merge and design a collaborative book chapter involving contributions from 31 participants.

Digital Literacy Program Highlights:

Research & Content Development:

- Compiling comprehensive references on Cyber Security and Semiconductor-War.
- Gathering information on account cracking techniques to better educate against them.
- Developing engaging Cyber Security socialization materials for students.

Planning for Technical Demonstrations:

- Designing and preparing a website functionality simulation to illustrate digital concepts.
- Developing and preparing materials for a smart home demonstration, showcasing how relays and dot matrices can be controlled.
- Assembling microcontroller equipment, relays, and dot matrices to support the smart home demonstration.
- Setting up equipment for running text displays.

CERTIFICATION

Information and Communication Technology (ICT)

- Mikrotik: MTCNA – Jan 2019
- Cisco: Network Automation Engineering Fundamentals Specialization, Endpoints and Systems (Coursera)- March 2024
- Cisco CyberOps Associate (CCOA) - May 2024
- JavaScript Beginner (Udemy and Skilvul)- March 2024

- Basic (Python, SQL), Intermediate (SQL) HackerRank- Feb 2024

- Pure Storage Field Engineer, FlashBlade and FlashArray

English Course

- ETS TOEIC score 570- Feb 2019
- Centralcourse TOEFL score 553- Sep 2023

SKILLS

I have broad experience across multiple IT fields, including system integration, troubleshooting, automation, and web technologies. I enjoy adapting to new challenges and building efficient solutions.

TOOLS & TECHNICAL SKILLS

Languages & Frameworks:

- Python (Flask, Django), C++, Java, GoLang, JavaScript (Node JS, Express, VueJS, NextJS), Tailwind, Bootstrap.

Infrastructure & Networking:

- VMware, Proxmox, Citrix, NAT, VPN/IPsec, Policy-Based Routing (PBR), Network Troubleshooting, Wireless Networking, DDoS Analysis

Tools:

- Wireshark, Kibana, Git, SIM800L, MQTT, AutoCAD, SketchUp, Blender